

Algebraic Topology: a beginner's course

This lecture:

**AlgTop33: Computing
homology groups**

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Lecture 33 Computing homology groups.



Lecture 33 Computing homology groups

pace

$$\begin{array}{ccccccc} & & \partial_3 & & \partial_2 & & \partial_1 & & \partial_0 = 0 \\ C_3 & \longrightarrow & C_2 & \longrightarrow & C_1 & \longrightarrow & C_0 & \longrightarrow & 0 \\ \text{3-dim} & & \text{2-dim} & & \text{1-dim} & & \text{0-dim} & & \\ \text{chains} & & \text{chains} & & \text{chains} & & \text{chains} & & \end{array}$$

Some examples

Ex 1 $X = S^1$



$\approx$



skeleton of a Δ :
3 line segments.



Some examples

Ex 1 $X = S^1$ 



skeleton of a Δ :

3 line segments

vertices $\{x, y, z\}$

3 1-simplices (segments)

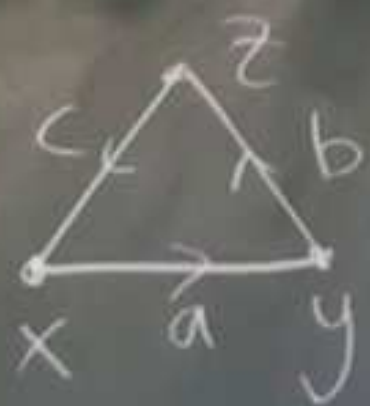
$a = (xy), b = (yz), c = (zx)$

examples

$$X = S^1$$



$\approx$



skeleton of a Δ :
3 line segments.

vertices $\{x, y, z\}$
3 1-simplices (segments)

$$X = \{a, b, c, x, y, z\}$$

$$Z_0 = \ker \partial_0 = C_0 = \langle x, y, z \rangle$$

$$B_0 = \text{im } \partial_1 = \langle y-x, z-y, x-z \rangle$$

$$a = (xy), b = (yz), c = (zx)$$

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0=0} 0$$

0	$\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$	$\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$
	$a \quad b \quad c$	$x \quad y \quad z$
	$\alpha a + \beta b + \gamma c$	$\underbrace{\hspace{1cm}}$
		$\alpha x + \beta y + \gamma z$

$$\partial_0 = 0: x, y, z \rightarrow 0$$

$$\partial_1: \begin{aligned} a &\rightarrow y-x \\ b &\rightarrow z-y \\ c &\rightarrow x-z \end{aligned}$$

$$\partial_2: 0 \rightarrow$$

Some examples

$X = S^1$
 $Z_0 = \ker \partial_1$



skeleton of a Δ :
 3 line segments

vertices $\{x, y, z\}$
 3 1-simplices (segments)

$X = \{a, b, c, x, y, z\}$

$a = (xy), b = (yz), c = (zx)$

$\partial_0 = 0: x, y, z \rightarrow 0$
 $\partial_1: \begin{matrix} a \rightarrow y-x \\ b \rightarrow z-y \\ c \rightarrow x-z \end{matrix}$ B_1

$la + mb + nc$
 $(y-x)$

$h(z-y)$
 $h(x-z)$

$Z_0 = \ker \partial_0 = C_0 = \langle x, y, z \rangle$
 $B_0 = \text{Im } \partial_1 = \langle y-x, z-y, x-z \rangle$

$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$

$\begin{matrix} \mathbb{Z} & \mathbb{Z} & \mathbb{Z} & \mathbb{Z} \\ a & b & c & \\ \hline \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} \\ x & y & z \\ \hline la + mb + nc \\ \hline \alpha x + \beta y + \gamma z \end{matrix}$

$(n-l)x + (l-m)y + (m-n)z = 0 \Rightarrow l=m=n$ so $\ker \partial_1 = \langle a+b+c \rangle = Z_1$

Q What does $H_0(X)$ measure?

A The # of connected components

$$C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

$$Z_0 = C_0 = \ker \partial_0$$

$$B_0 = \text{im } \partial_1 = \langle x_2 - x_1, x_3 - x_2 \rangle$$

Equating all elements of $B_0 \equiv 0 \Rightarrow$

Setting all vertices equal to each other.

So $x_1 = x_2 = x_3$ quotient x_1

$x_1, x_2, x_3 \in X$

$\langle x_1, x_2, x_3 \rangle \subset X$



$\langle x_1, x_2, x_3 \rangle$

$\langle x_1, x_2, x_3 \rangle$

